AeroGuardian: Proactive UAV Safety Intelligence System

Transforming FAA UAS sighting reports into PX4 simulations for pre-flight safety intelligence

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OVERVIEW

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CHAPTER 1

Introduction

=> Generative AI + X (Aerospace Safety Engineering)

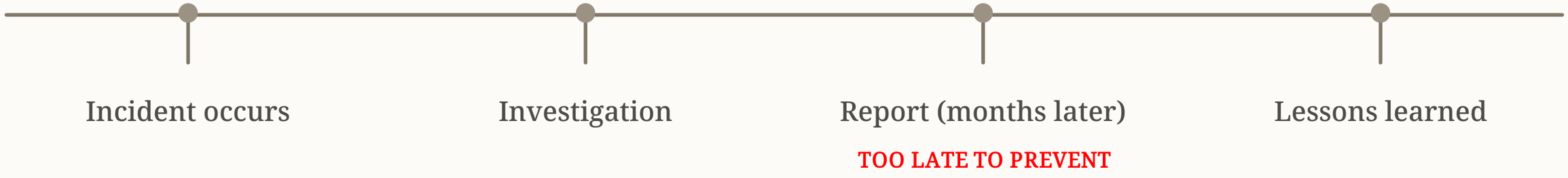
Component	Description
AI	Generative AI (GPT-4o)
X	Aerospace Safety / UAV Flight Safety
Fusion	LLM-driven scenario translation + Physics simulation

Background

- 8,031 FAA UAS sightings (2019-2025) — unanalyzed safety incidents.
- Growing UAV market — 2.4M+ registered drones in US alone.
- Safety gap — **No proactive** pre-flight analysis from incident data.

CHAPTER 2

The Problem Definition



Issue	Impact
8,000+ FAA reports	Impossible to analyze manually
Unstructured text	No systematic parameter extraction
No simulation testing	Cannot validate safety hypotheses
Time-consuming	7+ hours per incident analysis

CHAPTER 2

Problem Definition - Gap in Existing Technologies

Technology	What It Does	Why It's NOT Enough
CNN/YOLO	Detect drones in images	Cannot predict failures before flight
ML anomaly detection	Find anomalies in data	Requires past flight data (**reactive**)
Rule-based systems	Apply fixed rules	Cannot understand natural language
Flight recorders	Record flight data	Only works AFTER incident
Manual analysis	Human reviews reports	Too slow, not scalable

CHAPTER 3

Proposed Solution

> We propose **AeroGuardian**, an AI-based pre-flight safety analysis framework where Generative AI (LLMs) acts as a reasoning agent rather than a simple text generator.

Goal

Systematically transform real-world **abnormal flight reports** into **executable safety scenarios** to evaluate mission risk and produce kind of **actionable recommendation** before flight.

Key Innovation

It automatically links real-world **abnormal flight reports** knowledge to simulation-based testing, **providing explainable**, **data-driven safety decisions** and **human-readable reports** for operators.

CHAPTER 4

Methodology - System Architecture

1

Data Collection

Curating unstructured narratives from **FAA** abnormal and near-miss reports **(2019–2025)**

2

NLP Translation using **LLM**

Utilizing **GPT-4o** to perform interpretation, Extracting **31 typed simulation parameters** from raw text.

3

Simulation Execution

Executing scenarios in **PX4 SITL** with automated **fault injection** for propulsion, navigation, and sensor failures.

4

Telemetry Capturing and Analyzing

Capturing high-frequency telemetry such as **position drift, velocity spikes, and etc. Deterministic threshold detection and Causal chain analysis**.

5

Safety Synthesis using LLM

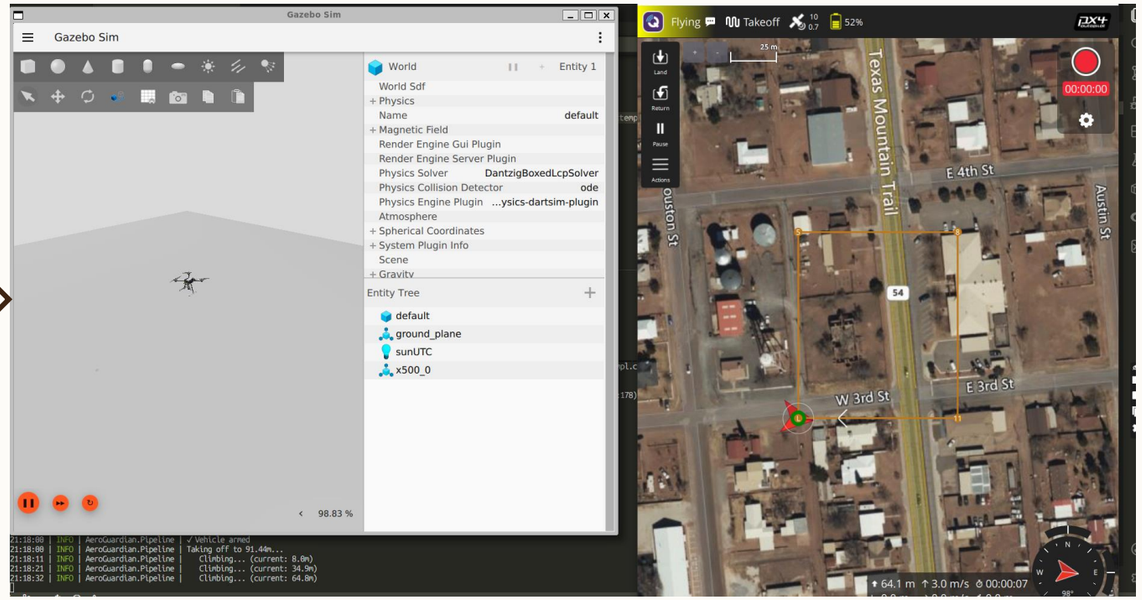
Generating reports with quantified **safety Level & cause**, **recommendations**, **evidence-based explanation**.

Methodology - Pipeline Results (Step by Step)


```
{
    "report_id": "DEMO_MOTOR_001",
    "date": "2020-04-10",
    "city": "VAN HORN",
    "state": "TEXAS",
    "description": "UAS malfunctioned during flight operations....
}
```



```
"llm_configuration": {
  "mission": {
    "start_lat": 31.0412,
    "start_lon": -104.8308,
    "takeoff_altitude_m": 91.44,
    "flight_mode": "MISSION",
    "duration_sec": 120,
    "cruise_speed_ms": 10.0,
    "cruise_altitude_m": 91.44
  },
  "fault_injection": {...
  },
  "environment": {...
  },
  "gps": {...
  },
  "waypoints": [...
  ],
  "px4_commands": {
```

PRE-FLIGHT SAFETY REPORT

Report ID: DEMO_MOTOR_001
Location: VAN HORN, TEXAS

1. SAFETY LEVEL & ROOT CAUSE

Safety Level: CRITICAL
Pre-Flight Decision: NO-GO
Root Cause Subsystem: CONTROL (confidence: 88%)
Causal Chain: control -> navigation
Primary Hazard: Simulated: Motor failure scenario producing loss of control
Observed Effect: In simulation: Motor failure produced 180-degree roll and pitch excursions, leading to significant altitude and position deviations

2. DESIGN CONSTRAINTS & RECOMMENDATIONS

Design Constraints:
- Consider: Maintain minimum altitude of 30m AGL for recovery time
- Consider: Limit operations over populated areas
- Consider: Visual line of sight operations

Recommendations:
1. Consider: Redundant motor configuration may reduce single-point failure risk
2. Consider: Pre-flight vibration monitoring
3. Consider: Parachute recovery system
4. Consider: Regular motor maintenance checks
5. Consider: Enhanced flight control algorithms to manage motor failure

3. EXPLANATION

This analysis simulated a motor failure scenario. In simulation, the fault produced extreme roll and pitch deviations, indicating a loss of control. NOTE: This represents quadrotor dynamics and may not match the reported aircraft. The critical severity of the anomalies suggests a high risk of crash without intervention. Recommendations are suggested mitigations for similar scenarios, including redundant systems and enhanced control algorithms.

Generated: 2026-02-07 21:19 | AeroGuardian v1.0 Pre-Flight Safety System

Analyzing the captured telemetry against the threshold that we tested with real data of RflyMAD Dataset (Beihang University) before feeding to LLM.


```
"telemetry": [
  {...
  },
  {
    "timestamp": 0.07941102981567383,
    "lat": 31.042972,
    "lon": -104.83162779999999,
    "alt": 0.21900001168251038,
    "relative_alt": -0.009000000543892384,
    "roll": -0.08539464324712753,
    "pitch": -0.21959948539733887,
    "yaw": 98.77750396728516,
    "actuator_controls_0": [...
    ],
    "acc_x_m_s2": -0.01992193609476089 5,
    "acc_y_m_s2": 0.01860857382416725,
    "acc_z_m_s2": -9.80203628540039,
    "gyro_x_rad_s": 0.0002464214339852333,
    "gyro_y_rad_s": -6.076046702219173e-05,
    "gyro_z_rad_s": -0.0009795159567147493
  },
```

CHAPTER 4

Methodology - Quantitative Validation

Benchmark: RflyMAD Dataset (Beihang University)

Metric	Value
Total Flights	1,424
Total Samples	**1,418,960**
Precision	**92.9%**
Recall	63.4%
F1-Score	**75.3%**

Per-Fault Performance

Fault Type	F1-Score
Motor Fault	**87.7%**
Sensor Fault	86.1%
Wind Fault	53.4%

CHAPTER 4

Methodology - Comparison with Recent Research

System	Pre-flight?	NL Input?	Accuracy
SOD-YOLO	✗	✗	94% mAP
1DCNN-SVDD	✗	✗	90% AUC
RADD	✗	✗	High
AeroGuardian	✓	✓	**75.3% F1**

CHAPTER 4

Methodology – Time Efficiency

Task	Manual	AeroGuardian	Savings
Read report	30 min	2 sec	99%
Extract parameters	1 hour	5 sec	99%
Simulation	N/A	90 sec	New
Analyze telemetry	2 hours	0.1 sec	99%
Generate report	4 hours	30 sec	99%
TOTAL	**7+ hours**	**~3 min**	**99%**

CHAPTER 5

Demonstration

The video is playing...

CHAPTER 6

Conclusion: Advancing UAV Safety

- **Operationalizing Data**

  Our approach introduces a novel method to operationalize historical aviation safety data.

- **Dynamic Simulations**

  It transforms static FAA narratives into executable, physics-based simulations.

- **Proactive Safety**

  Enables proactive, data-driven pre-flight safety analysis, shifting from reactive to predictive safety.

- **Scalable Benchmarking**

  Provides a robust foundation for scalable safety benchmarking and future certification-oriented tools.

- **LLMs for Safety**

  Positions LLMs as powerful semantic-to-physical translators for critical aviation safety applications.

Thank You!

Team Tiny Coders

Github: https://github.com/rak-junior/aero-guardian.git